Regression and Residuals

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Match the quantities. Declare the sign. Keep the magnitudes.

Interpret numerical prediction errors without confusing cancellation with accuracy.

Keep one habit: Complete three-case example → signed residual bars → six-case Residual Walk → missing/extreme probes → independent new pairs.

MAIN LESSON · SECTION 1

How far and in which direction did the estimate miss?

> Residual signs preserve direction; absolute error preserves size.

Learning goals

I can distinguish numeric **prediction** from classification.

I can calculate signed **residual**s with a stated convention.

I can use **residual** patterns to question a model.

Retrieve target meaning and error evidence

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Accuracy and Imbalance. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.
- What does a class label predict?

End goal

By the end, I can explain **regression** and **residual**s using a traced example and a stated limitation.

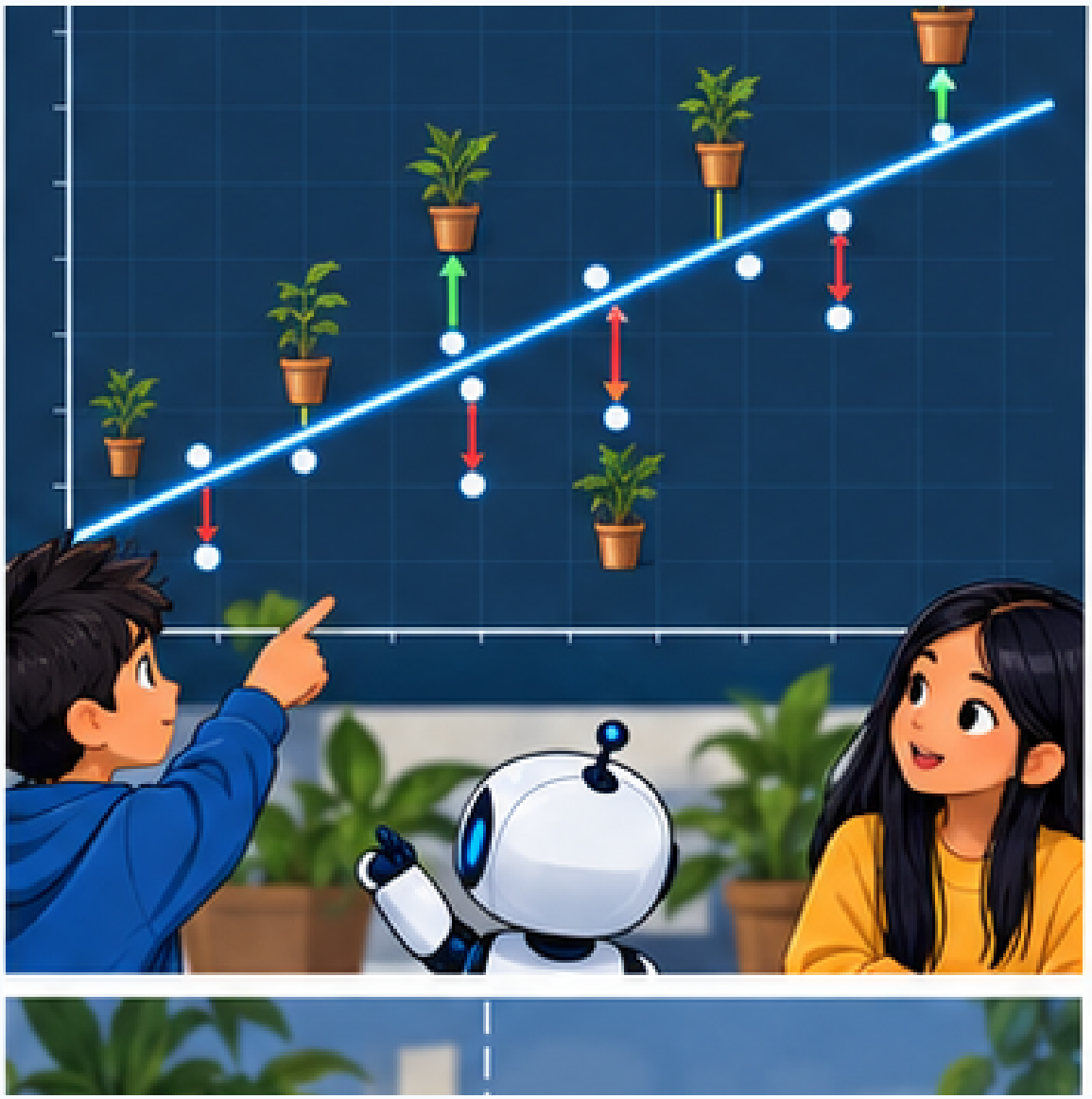
▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Regression predicts a number; a **residual** records signed error and reveals direction as well as size.



LEARNING TARGET Actual leaf length is 7 cm and **prediction** is 10 cm. Using **residual**=actual−predicted, calculate **residual** and absolute error. Interpret the sign.

Builder name and experiment date:

Readiness for the connected explanation

This week’s end goal

Define the quantity, unit and subtraction order

> Under observed-minus-predicted, a positive residual means the prediction was too low.

Numerical prediction and sign contract

A school garden example predicts plant height in cm from recorded conditions. The target is numeric, so a **prediction** can be too high or too low by different amounts. Define **residual** as observed minus predicted.

Keep units and the sign convention visible; some software uses the opposite error convention.

Here regression means prediction of a numerical quantity. In last week's software-change record, a regression meant a new or worsened failure; those uses of the word are different. These height examples use supplied predictions, not a newly trained model. Observe and predict the same case, quantity, time and unit before subtracting. A day-7 height is not a growth increment without an initial-height observation.

Fictional example scope

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

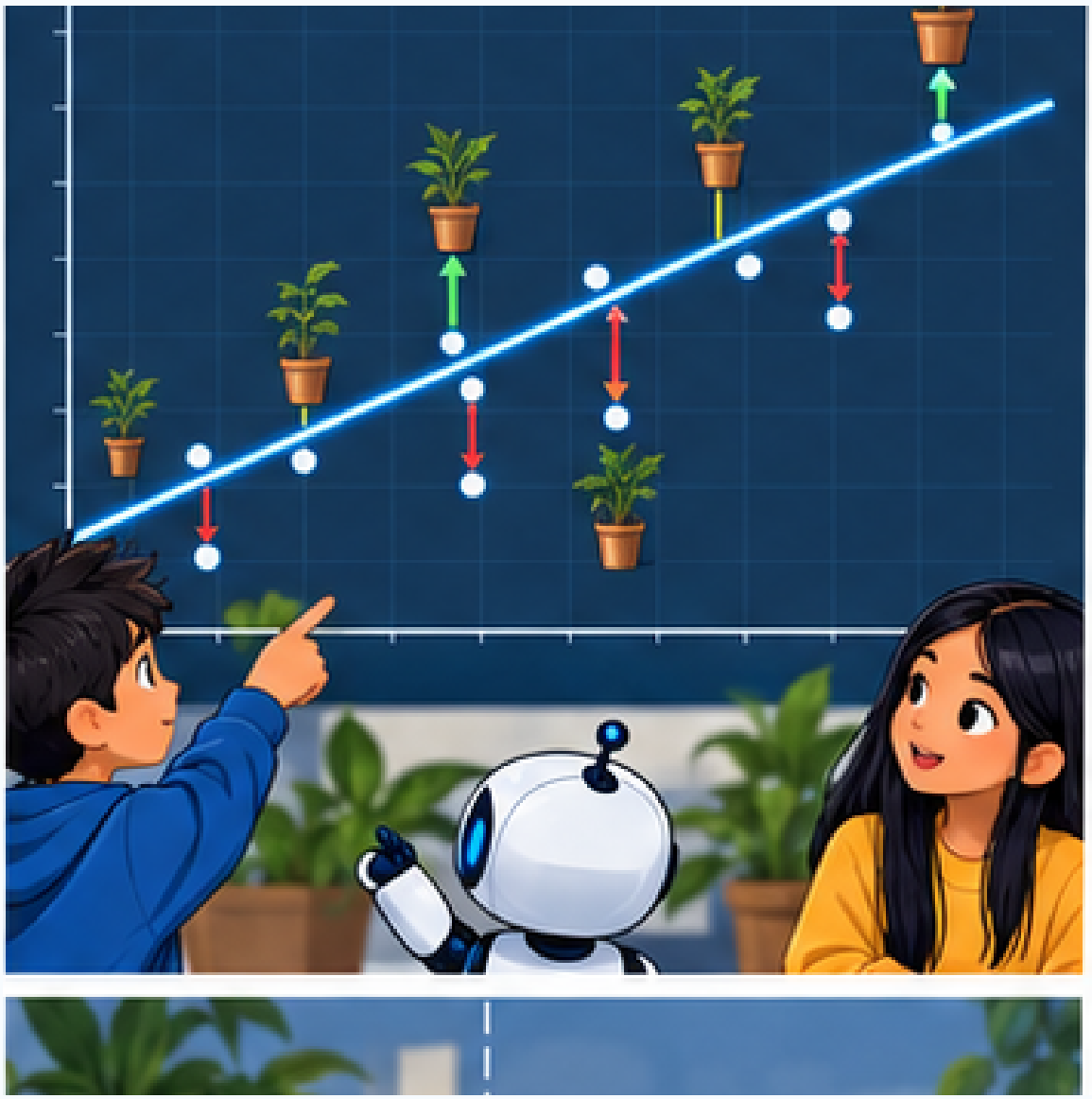
Earlier weeks supplied the model-building **process**; this week examines **regression** and **residual**s.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use **regression** and **residual**s to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

MAIN LESSON · SECTION 3

Calculate every matched residual before averaging

> Zero mean signed error can conceal several wrong predictions.

Complete three-case cancellation example

Observed heights [4, 7, 9] cm and **prediction**s [5, 6, 9] cm give **residual**s [-1, 1, 0] cm. The first **prediction** is too high, the second too low and the third exact. Mean signed **residual** is zero, yet two **prediction**s are wrong.

Opposite errors can cancel.

Optional original calculation script

```
observed = [4, 7, 9]
predicted = [5, 6, 9]
residuals = [y - p for y, p in zip(observed, predicted)]
print(residuals)
print(sum(residuals) / len(residuals))
# Output: [-1, 1, 0] and 0.0.
# Opposite signed errors can cancel.
```

Trace and state matching assumptions

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

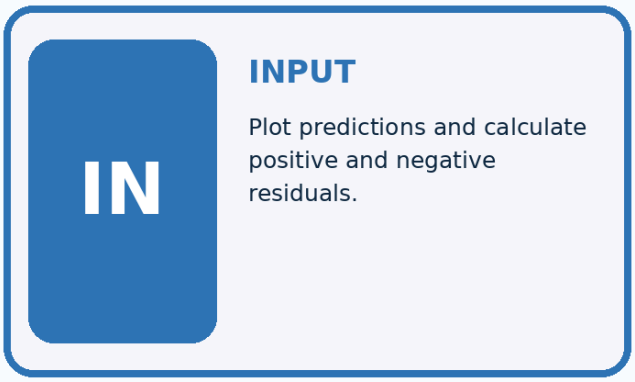
The original zip-based example assumes nonempty lists with equal lengths, matching case order and consistent height units. zip alone does not verify those conditions and silently truncates unequal lists. An exact residual0establishes equality only for that recorded pair, not a reliable model. Mean signed residual0can arise through cancellation; retain per-case magnitudes and sample size.

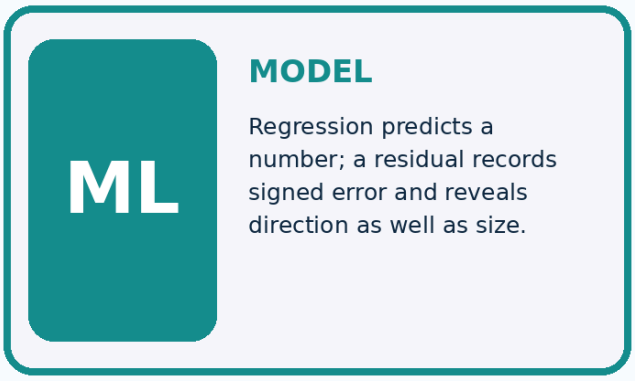
▼ Earlier wording, alternatives and picture transcripts

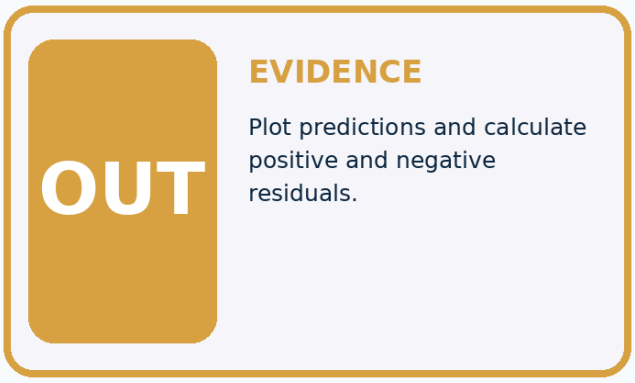
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION Plot **prediction**s and calculate positive and negative **residual**s.

MODEL IDEA **Regression** predicts a number; a **residual** records signed error and reveals direction as well as size.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Actual plant height is 30 cm; **prediction** is 27 cm. **Residual** = actual – predicted = 30 – 27 = +3 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Connect the picture and the explanation

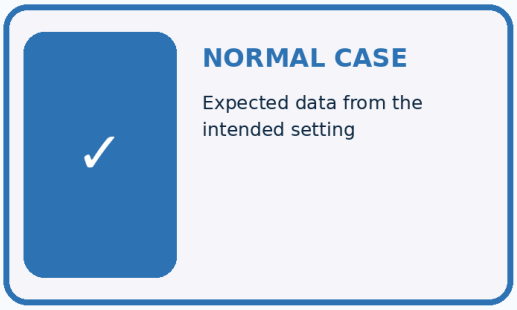
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

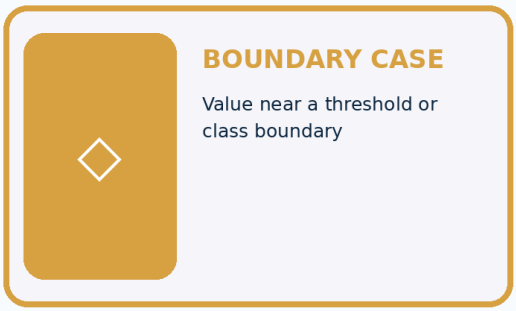
Compare underprediction, overprediction and missing observation

> The sign depends on a declared convention, not a memorized direction word.

Four evaluation conditions



NORMAL Expected data and expected behavior.

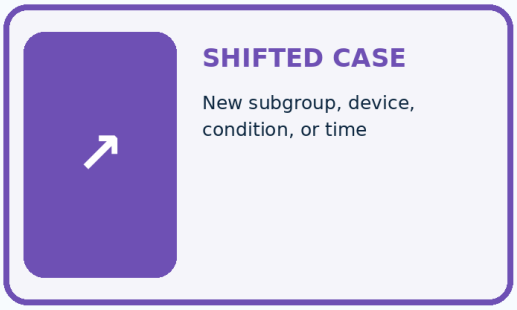


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: Raise all **prediction**s by1 cm.

▼ **Check after predicting**

Prediction: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.

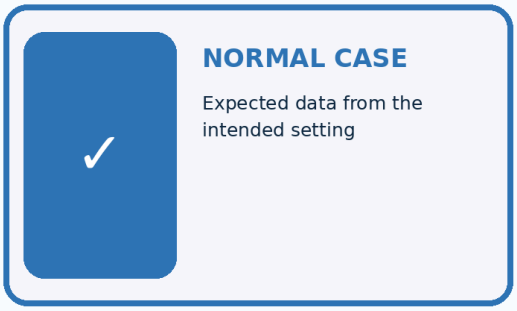
Reason: Each **residual** decreases by1 under the observed-minus-predicted convention.

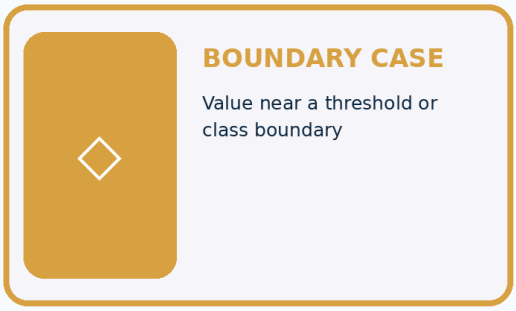
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

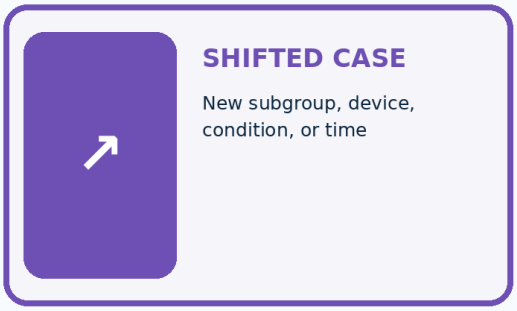
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Read signed bars and magnitudes from the same pairs

> Negative residual means the supplied prediction exceeded the observation.

Complete pictured values and scenario scope

Use **residual** signs to say whether each plant-height estimate is too high or low.

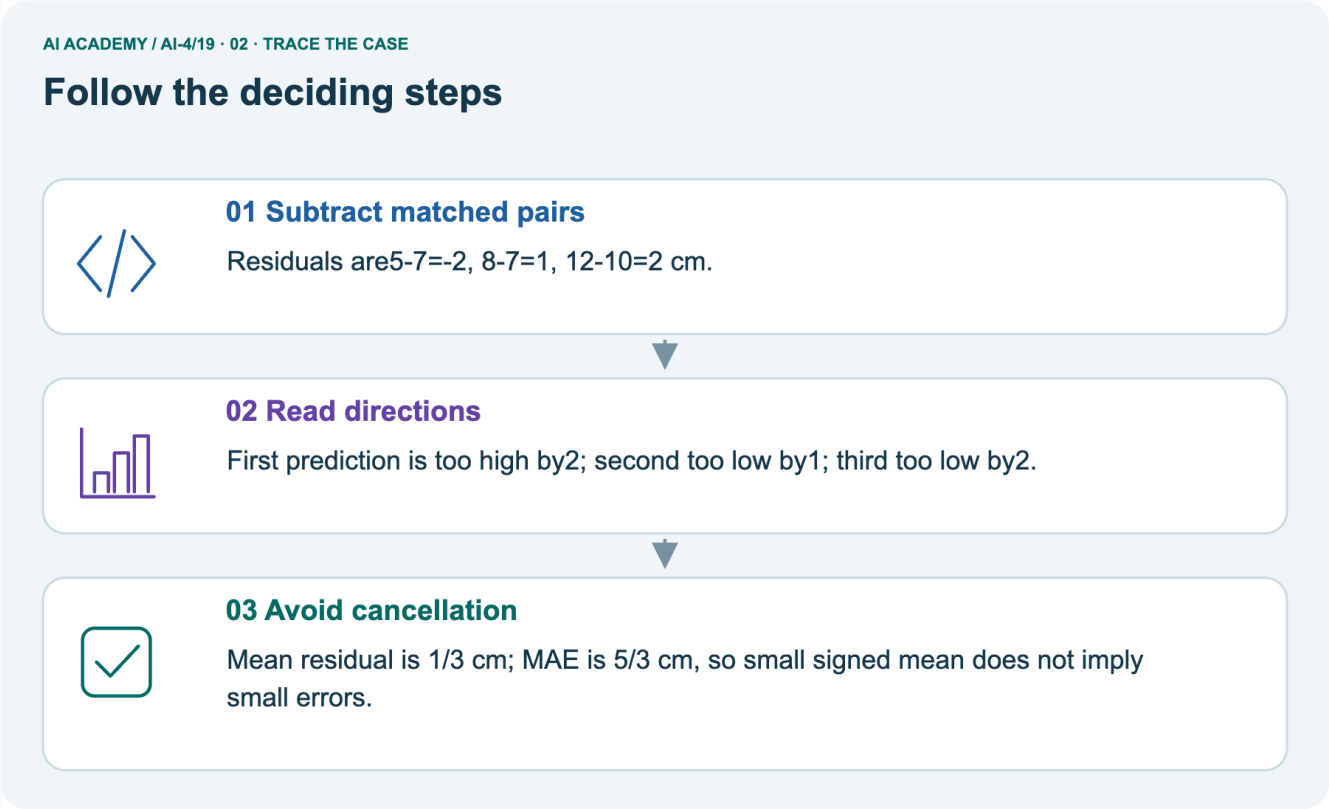
Residual=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main observed4,7,9versus predicted5,6,9 gives residuals-1,+1,0cm,mean0and MAE2/3cm. Picture observed5,8,12versus predicted7,7,10 gives-2,+1,+2cm,mean1/3and MAE5/3cm. Raising only the pictured predictions by1cm gives-3,0,+1,mean-2/3and MAE4/3cm. A smaller MAE on these already inspected records is a development observation,not proof of better future performance.

Keep observed values and the unit unchanged during this arithmetic comparison.

Trace signs, directions and cancellation



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

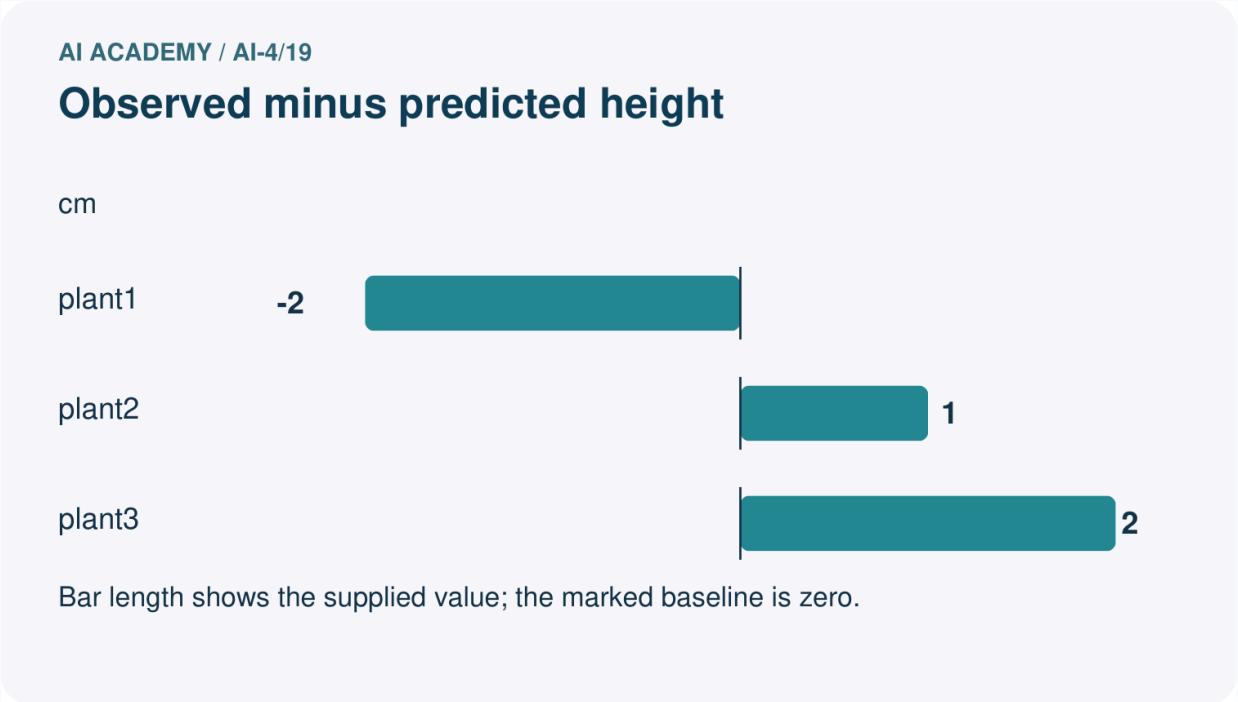
STEP	WHAT HAPPENS AND WHY
1. Subtract matched pairs	Residuals are5-7=-2, 8-7=1, 12-10=2 cm.
2. Read directions	First prediction is too high by2; second too low by1; third too low by2.
3. Avoid cancellation	Mean residual is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Inspect the original zero-centred residual bars

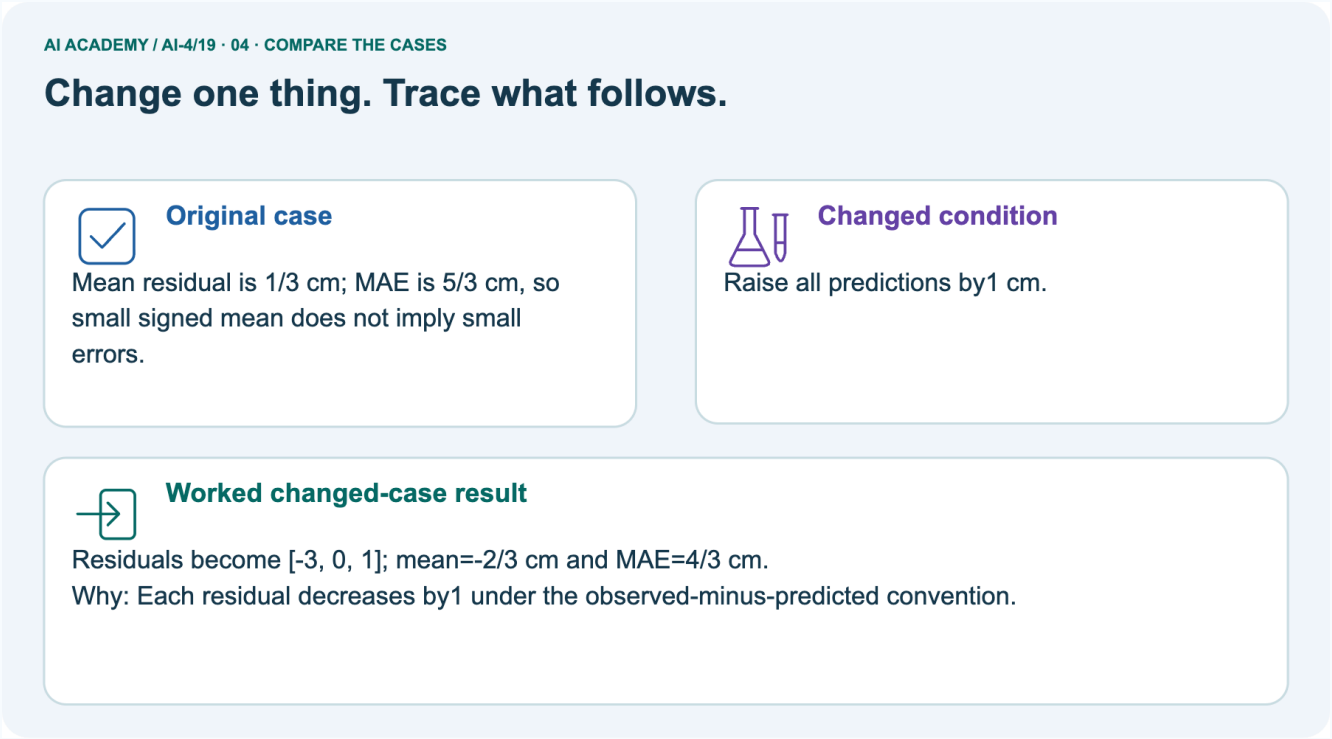


Follow the labelled picture with the matching explanation. Open the vector diagram

Predict raising every pictured prediction by1cm

- Change one thing: Raise all **prediction**s by1 cm. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain sign and error-size limits

- Always declare the **residual** sign convention; some software uses predicted minus observed.
- Read the labelled picture: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Return to the main cancellation question

Regression and residuals at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Observed 4 predicted 5	Residual minus 1	Overprediction
Observed 7 predicted 6	Residual plus 1	Underprediction
Observed 9 predicted 9	Residual zero	Exact on this record

To inspect a residual plot, place a feature or prediction on the horizontal axis and residual on the vertical axis. A trend can suggest a missed relationship. It does not by itself identify the cause or validate a new model. Use development data for analysis and preserve an independent final test.

Why does mean residual zero not imply perfect predictions? Explain the steps or evidence behind your response.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 19 • WORKED EVIDENCE

1. Actual plant height is 30 cm; prediction is 27 cm.
2. Residual = actual − predicted = 30 − 27 = +3 cm.
3. Another case has actual 20 and predicted 24, residual −4 cm.
4. Plot residuals; a pattern above or below zero suggests systematic model error.

STEP 1 Actual plant height is 30 cm; **prediction** is 27 cm.

STEP 2 **Residual** = actual – predicted = 30 – 27 = +3 cm.

STEP 3 Another case has actual 20 and predicted 24, **residual** –4 cm.

STEP 4 Plot **residual**s; a pattern above or below zero suggests systematic model error.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

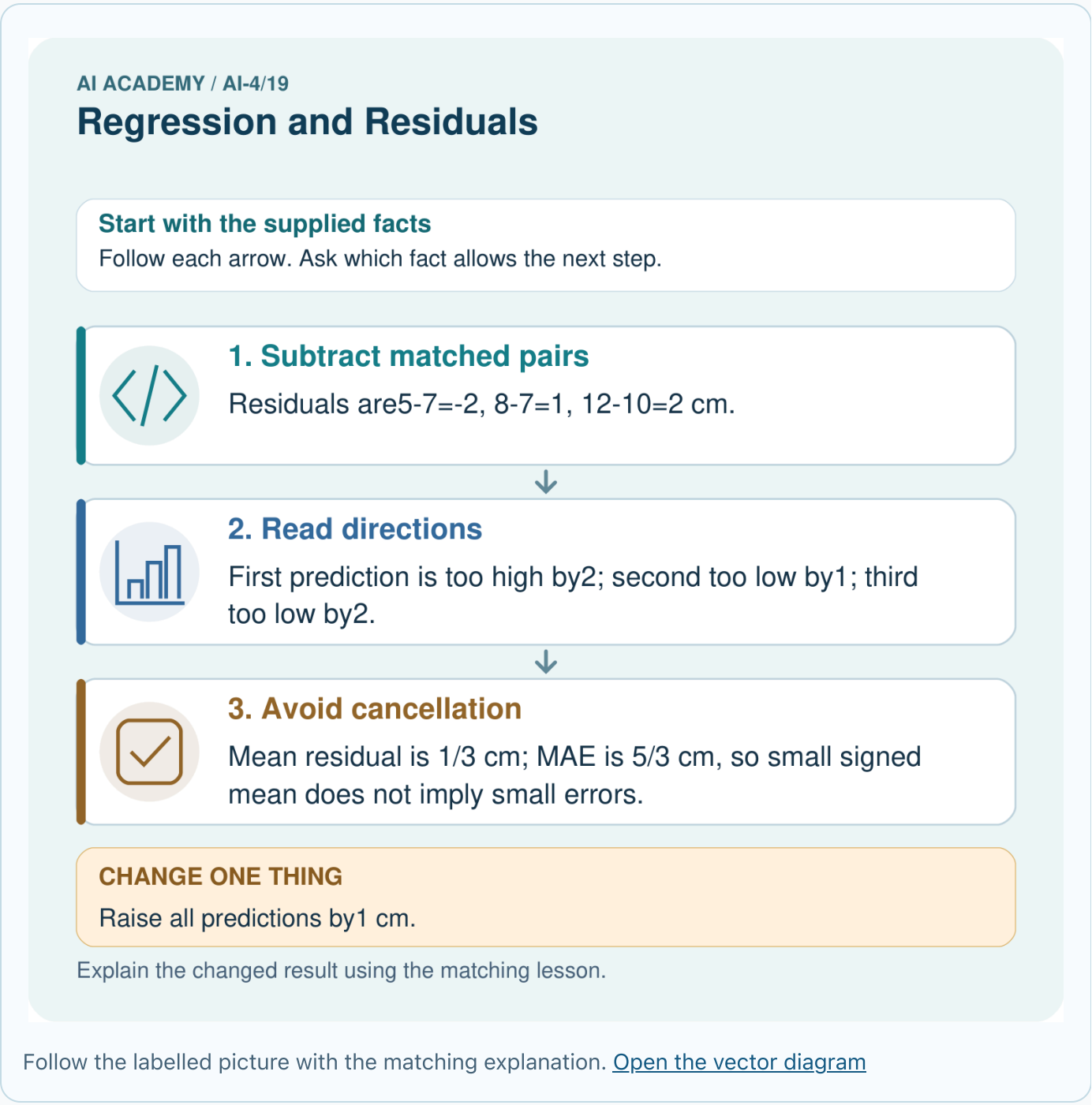
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Regression and Residuals

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a signed average mistaken for small error

> Preserve each residual before judging a model summary.

Diagnose the claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Positive and negative **residual**s cancel, so their average proves accuracy.

Explain the limitation and next check

- A **residual** of zero is an agreement with the measured target, not proof that the measurement is perfect. Do not compare errors measured in cm directly with errors measured in metres.

  Keep the same row pairing, unit and sign convention throughout.

> **A report combines 2 cm and .02 m as if they were different magnitudes. Repair it.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Positive and negative residuals cancel, so their average proves accuracy. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

A closer explanation

MAIN LESSON · SECTION 8

Practise cancellation and investigate a visible pattern

> Three points can motivate a check without establishing a population trend.

Try supported W2

Use workbook W2 for the connected example “**Regression** and **residual**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Inspect early/middle/late residual directions

If **prediction**s for early, middle and late growth have **residual**s -2, 0, 3 cm, the model moves from overpredicting to underpredicting across those records.

Investigate whether a straight-line assumption or a missing condition could explain it. Check measurement consistency before adding complexity.

Explain with a trace rather than a final number

- Use the worked case for **Regression** and **residual**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain which statistic retains magnitude

- Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

Reserve the new independent observed/predicted pairs

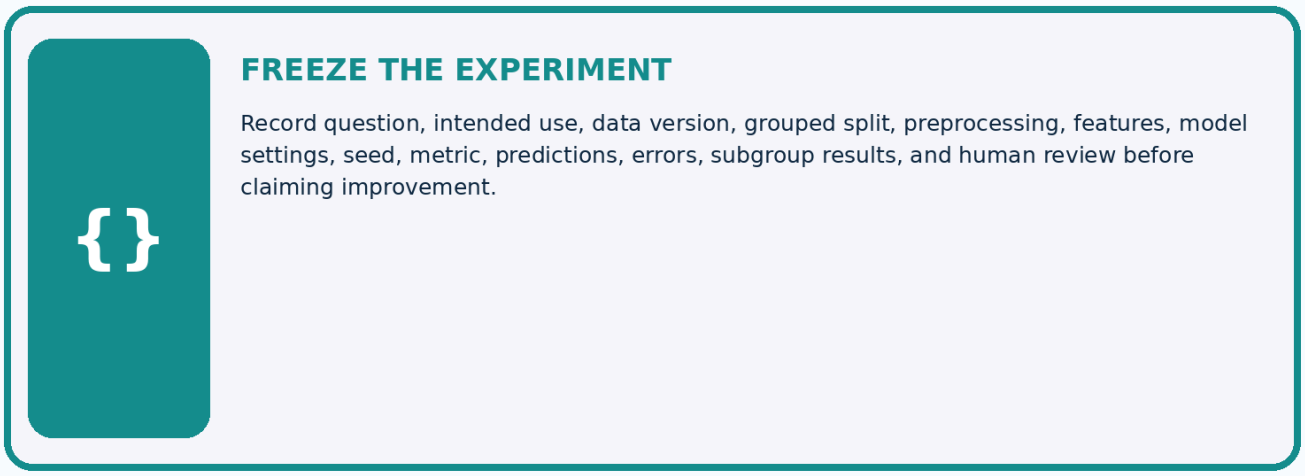
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Practise with less help: Explain cancellation

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run Residual Walk with complete matched measurements

> The supplied points support signs, cancellation and a bounded pattern discussion.

Complete six-case paired-height fixture

Case ID	Observed day-7 height cm	Supplied predicted height cm	Observed minus predicted cm
R1	3	4	-1
R2	5	4	1
R3	6	8	-2
R4	10	8	2
R5	8	12	-4
R6	16	12	4

Plot axes and checked summary

Residual Walk uses these six invented matched records. On a paper plot put supplied predicted height in cm horizontally and observed-minus-predicted residual in cm vertically, with a clearly marked zero line. Points are(4,-1),(4,1),(8,-2),(8,2),(12,-4),(12,4). Mean

residual0does not mean exact predictions: all six are wrong and MAE14/6=7/3cm.

This tiny constructed fixture has larger magnitudes at larger predicted heights; it suggests investigation,not a proved cause or stable population pattern.

Normal, equality, missing-observation and extreme-value probes

Complete residual probe	Expected bounded check
Normal R1 observed3/predicted4cm	Residual-1cm,overprediction by1cm,absolute error1cm.
Exact equality observed8/predicted8cm	Residual0and absolute error0for this pair.
Missing observed height for a proposed R7 prediction12cm	No residual is observed; retain unknown and exclude only under a documented summary policy. Do not replace it with0.
Separate extreme record R7 observed32/predicted12cm	Residual+20cm; investigate identity,measurement and unit. Do not delete it simply because it enlarges error.

Subtract, plot, explain and investigate

1. Predict all six residuals and place each point before checking the table or running the optional summary program. Say under/over/exact with the unit. Compare signed mean and mean absolute error; explain why cancellation occurs. For the separate extreme record, state source checks and a planned follow-up instead of automatically excluding it. Record missing observations and pairing assumptions.

   No regression model is fitted and no causal explanation for the constructed spread is supplied. Keep the selected three fresh observed/predicted pairs for an unaided first attempt.

Optional complete matched-record summary program

```
import math


def summarize(records):
```

```
    if not records:
        raise ValueError('No observed residual mean without matched records')
    if len({item_id for item_id, observed, predicted in records}) != len(records):
        raise ValueError('Repeated case ID')
    residuals = []
    for item_id, observed, predicted in records:
        for value in (observed, predicted):
            if type(value) not in (int, float) or not math.isfinite(value) or value < 0:
                raise ValueError('Need finite nonnegative heights in the same declared
unit')
        residuals.append(observed - predicted)
    return residuals, sum(residuals) / len(residuals), sum(abs(r) for r in residuals) /
len(residuals)


# Matched case ID, observed day-7 height cm, supplied predicted height cm.
records = [('R1', 3, 4), ('R2', 5, 4), ('R3', 6, 8),
           ('R4', 10, 8), ('R5', 8, 12), ('R6', 16, 12)]
print(summarize(records))
# Semantic pairing, measurement truth and units still require source checking.
```

Keep evidence and execution status

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Residual Walk: calculate signed errors, place them on a plot, and identify bias, outliers, or changing spread.

1. Write the hypothesis and expected **evidence** before running.

2. **Residual** Walk: calculate signed errors, place them on a plot, and identify bias, outliers, or changing spread.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: explain three fresh height residuals

> Show the matched subtraction and interpret its direction.

Complete independent values and scope

Observed [6, 10, 13], predicted [8, 9, 11] cm. Give **residual**s, mean **residual** and error directions.

Pair the new three observed and predicted heights in their supplied order, keeping cm throughout. Use observed minus predicted, then explain each sign before averaging. The independent pairs differ from the pictured values even if a resulting residual pattern happens to match; show subtraction evidence rather than copying a pattern. No fitting data or future-performance result is supplied.

First attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply independently

Week 19 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Define **residual** as actual minus predicted. Actual height is 14 cm and **prediction** is 17 cm. Calculate **residual** and interpret its sign.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Use a miniature growth-**prediction** side experiment to distinguish **regression evidence** from the main classifier metrics.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Actual leaf length is 7 cm and **prediction** is 10 cm. Using **residual**=actual−predicted, calculate **residual** and absolute error. Interpret the sign.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the regression side experiment's target explicit

> A numerical error in cm answers a different question from class-label accuracy.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Use a miniature growth-**prediction** side experiment to distinguish **regression evidence** from the main classifier metrics.

Keep the numerical side experiment separate from the classifier project metrics. Declare target quantity,time,units,source IDs and residual sign. Save observed/predicted pairs, missingness policy, per-case residuals and magnitude summary. Height is not growth without a stated baseline/time difference; a signed mean or reproducible plot does not establish generalization or measurement truth.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits, uncertainty and human decision

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

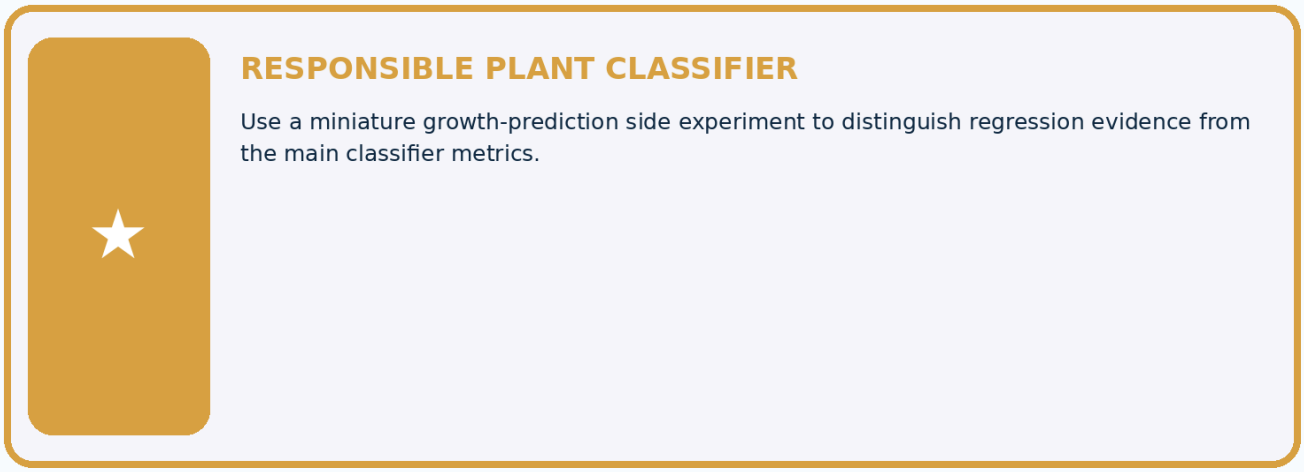
My **evidence**-linked project decision:

▼ **Earlier wording, alternatives and picture transcripts**

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with the sign, size and source pair

> Explain why zero mean signed residual can hide prediction errors.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Distinguish numeric **prediction** from classification

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
regression	prediction of a numerical target	
prediction	model-estimated value	
residual	actual value minus predicted value	
residual plot	errors displayed against predictions or inputs	

Closer-explanation terms

Word	Meaning
Regression	Prediction of a numeric target
Residual	Observed target minus predicted target in this course
Underprediction	A prediction below the observed value

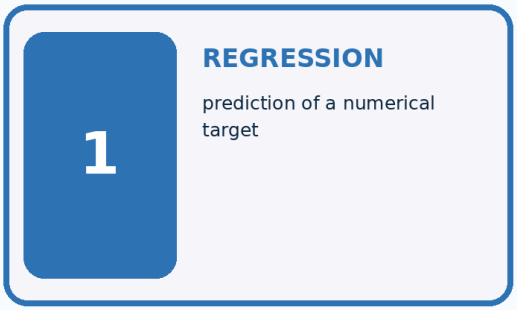
Word	Meaning
Overprediction	A prediction above the observed value

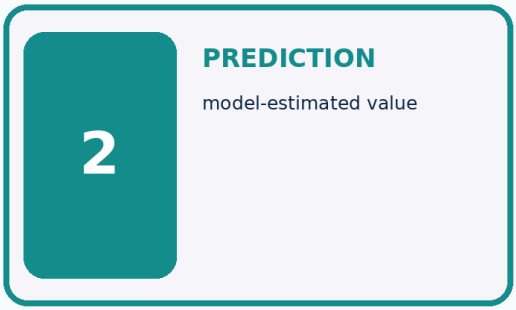
▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

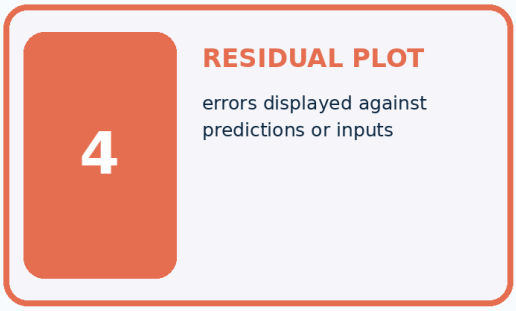
Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Positive and negative **residual**s cancel, so their average proves accuracy.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

▼ **Extra challenge · Build a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **regression**, **prediction**, **residual**, **residual plot**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Positive and negative **residual**s cancel, so their average proves accuracy.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

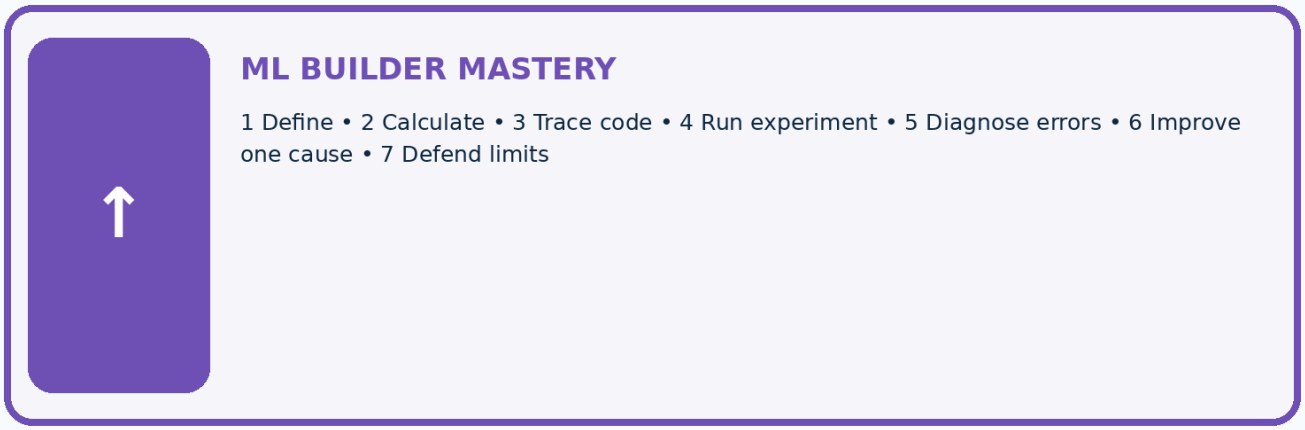
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh three-pair signed-residual task for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

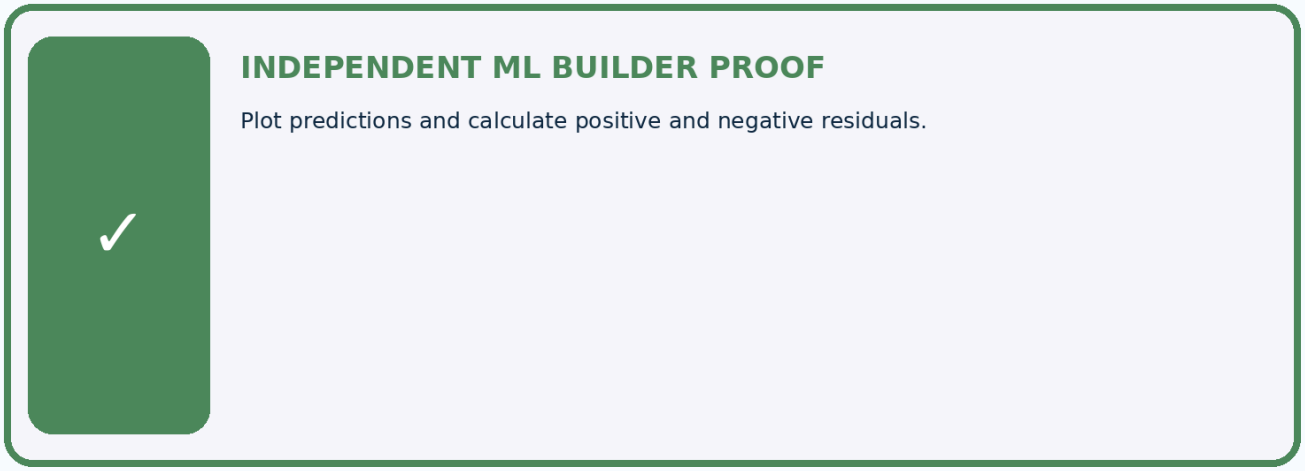
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF Plot **prediction**s and calculate positive and negative **residual**s.

NEXT BRIDGE Carry the experiment **evidence** into: Mean, Median, and Baselines.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Actual height=12 cm, **prediction**=9 cm. **Residual** means actual minus predicted.

Calculate and interpret.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

+3 cm means **underprediction** by 3 cm. State this sign convention consistently.

Evidence or stage	Value or result
Prediction	9
Actual	12

▼ Earlier wording, alternatives and picture transcripts

Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Residual is actual minus predicted

Actual height=12 cm, **prediction**=9 cm. **Residual** means actual minus predicted.

Calculate and interpret.

+3 cm means **underprediction** by 3 cm. State this sign convention consistently.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Distinguish numeric prediction from classification	Explain the deciding step, show a trace or test, and state one limit.
Calculate signed residuals with a stated convention	Explain the deciding step, show a trace or test, and state one limit.
Use residual patterns to question a model	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Use **residual** signs to say whether each plant-height estimate is too high or low.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Use **residual** signs to say whether each plant-height estimate is too high or low.

- Which labelled step matches this detail? Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.
- 1. Read directions
- 2. Avoid cancellation
- 3. Subtract matched pairs
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Avoid cancellation Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. Read the picture in order.

Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Raise all **prediction**s by1 cm. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Each **residual** decreases by1 under the observed-minus-predicted convention.

Repair and keep the record

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Accuracy and Imbalance (ai-4/16). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why can the 90% baseline still be unsuitable for a detection task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Train a Classifier (ai-4/13). Goal: Connect **training**-only scaling and neighbour **prediction** in one pipeline. Short practice: Which pipeline record would help another team reproduce the result?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

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<summary>Reference · The six-picture journey</summary>

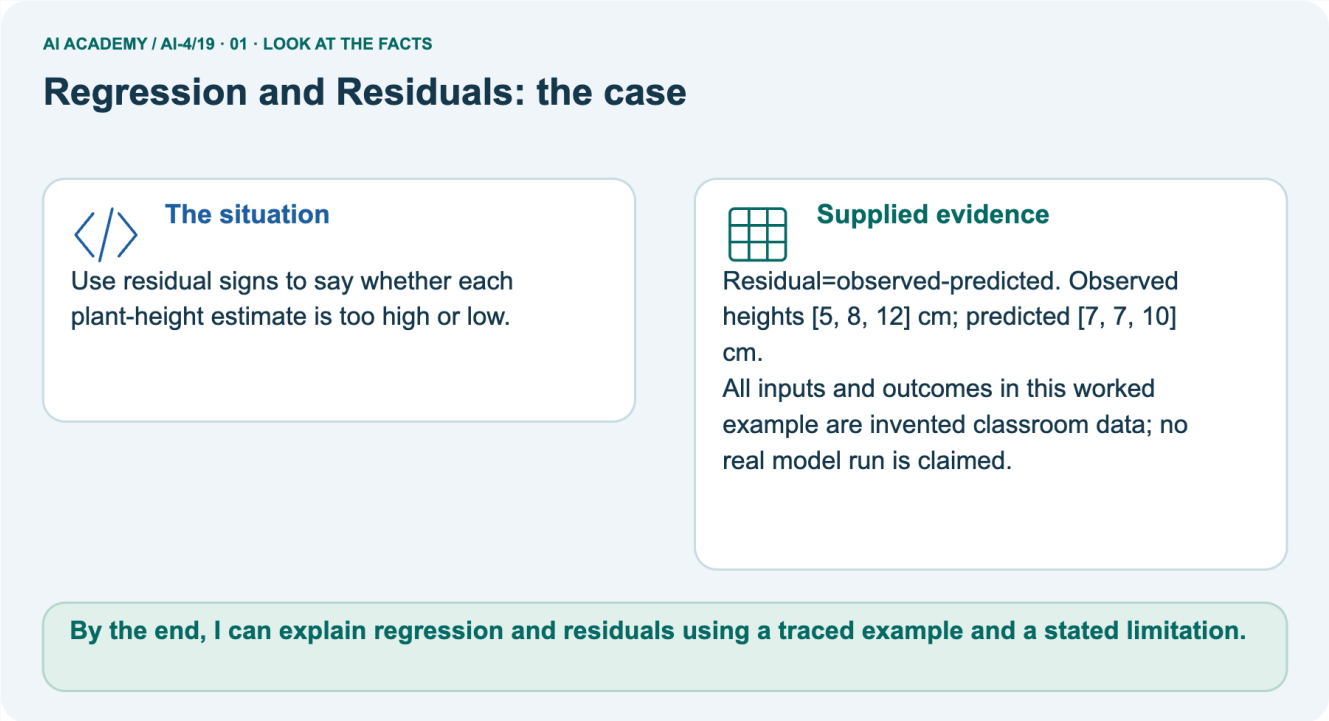
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Regression and Residuals: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Use **residual** signs to say whether each plant-height estimate is too high or low. Supplied facts: **Residual**=observed-predicted. Observed heights [5, 8, 12] cm; predicted [7, 7, 10] cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

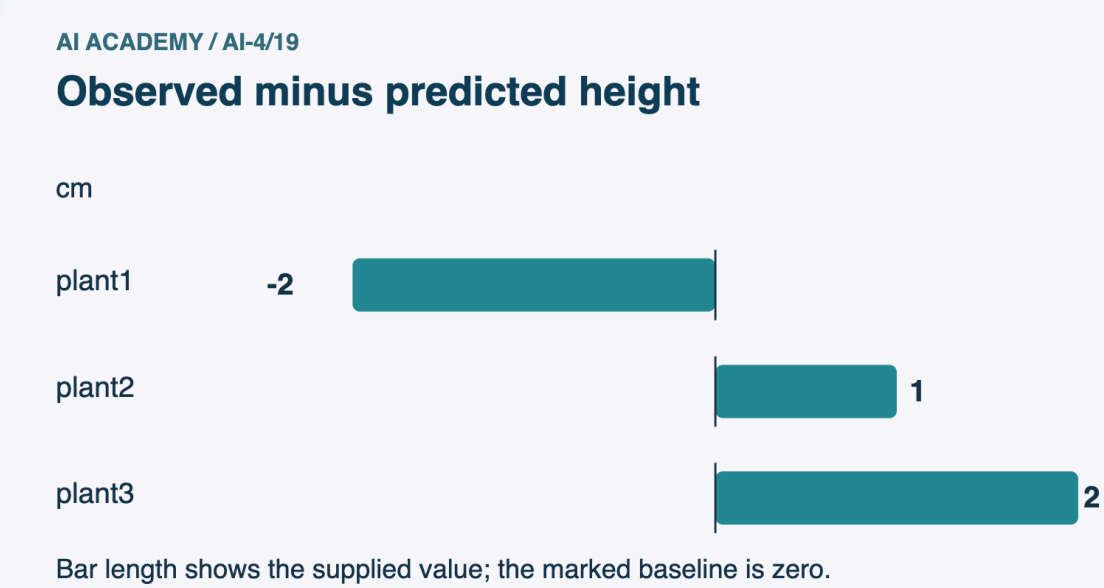


Follow the labelled picture with the matching explanation. Open the vector diagram

Subtract matched pairs: **Residual**s are5-7=-2, 8-7=1, 12-10=2 cm. → Read directions: First **prediction** is too high by2; second too low by1; third too low by2. → Avoid cancellation: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors.

Point to each numbered stage and explain which recorded **input** it uses.

03. Observed minus predicted height

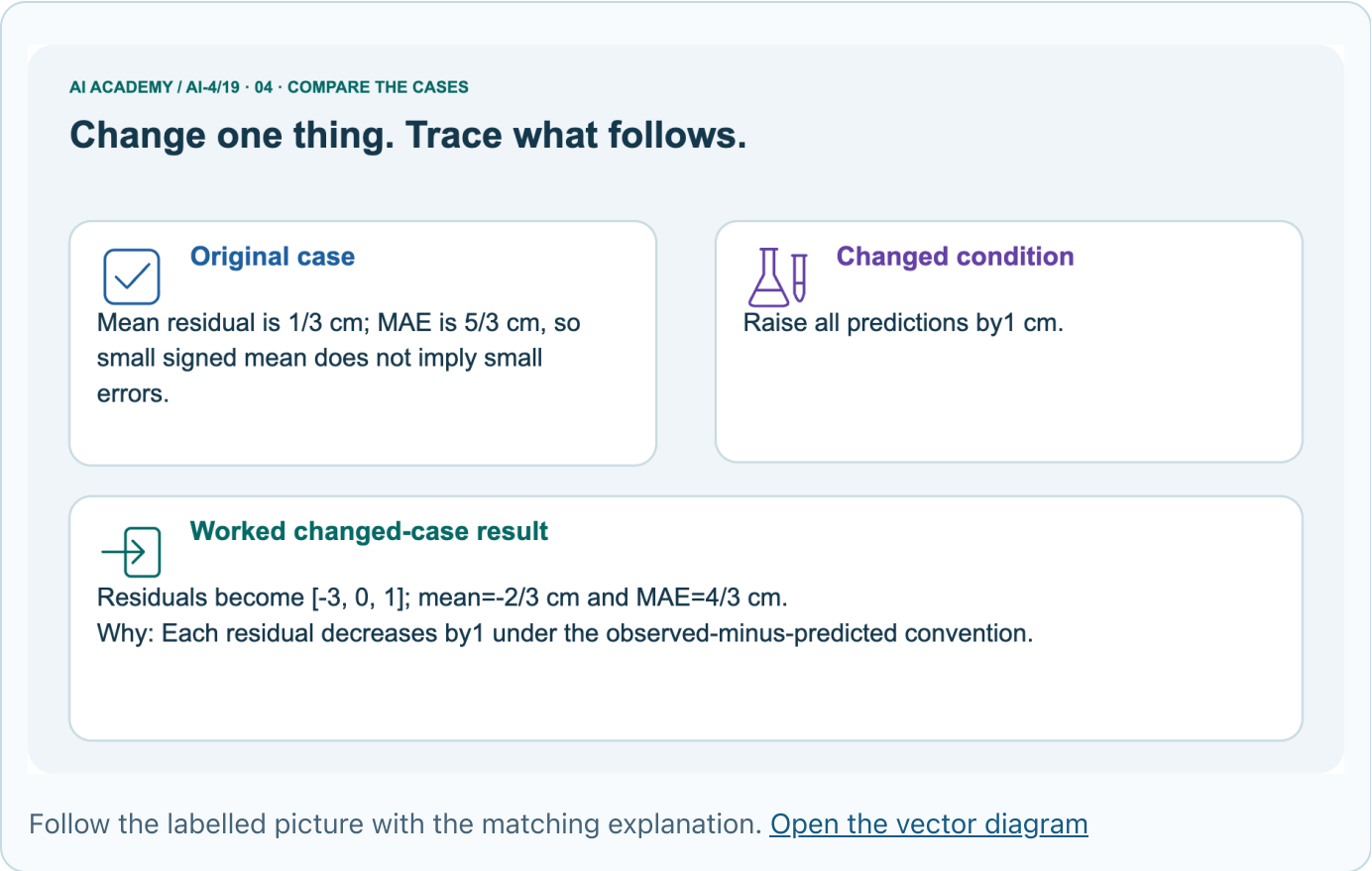


Follow the labelled picture with the matching explanation. Open the vector diagram

Observed minus predicted height. Units: cm. plant1: -2. plant2: 1. plant3: 2.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



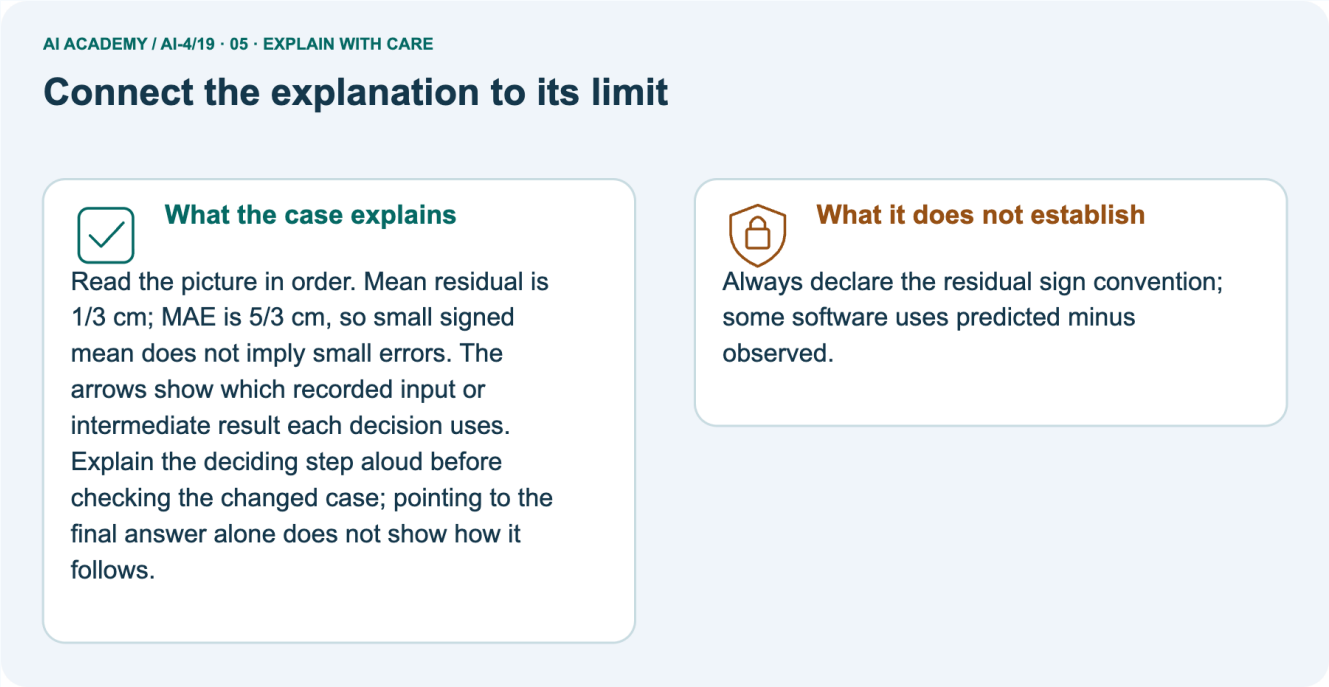
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. Changed condition: Raise all **prediction**s by1 cm.

Expected result: **Residual**s become [-3, 0, 1]; mean=-2/3 cm and MAE=4/3 cm. Reason: Each **residual** decreases by1 under the observed-minus-predicted convention.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

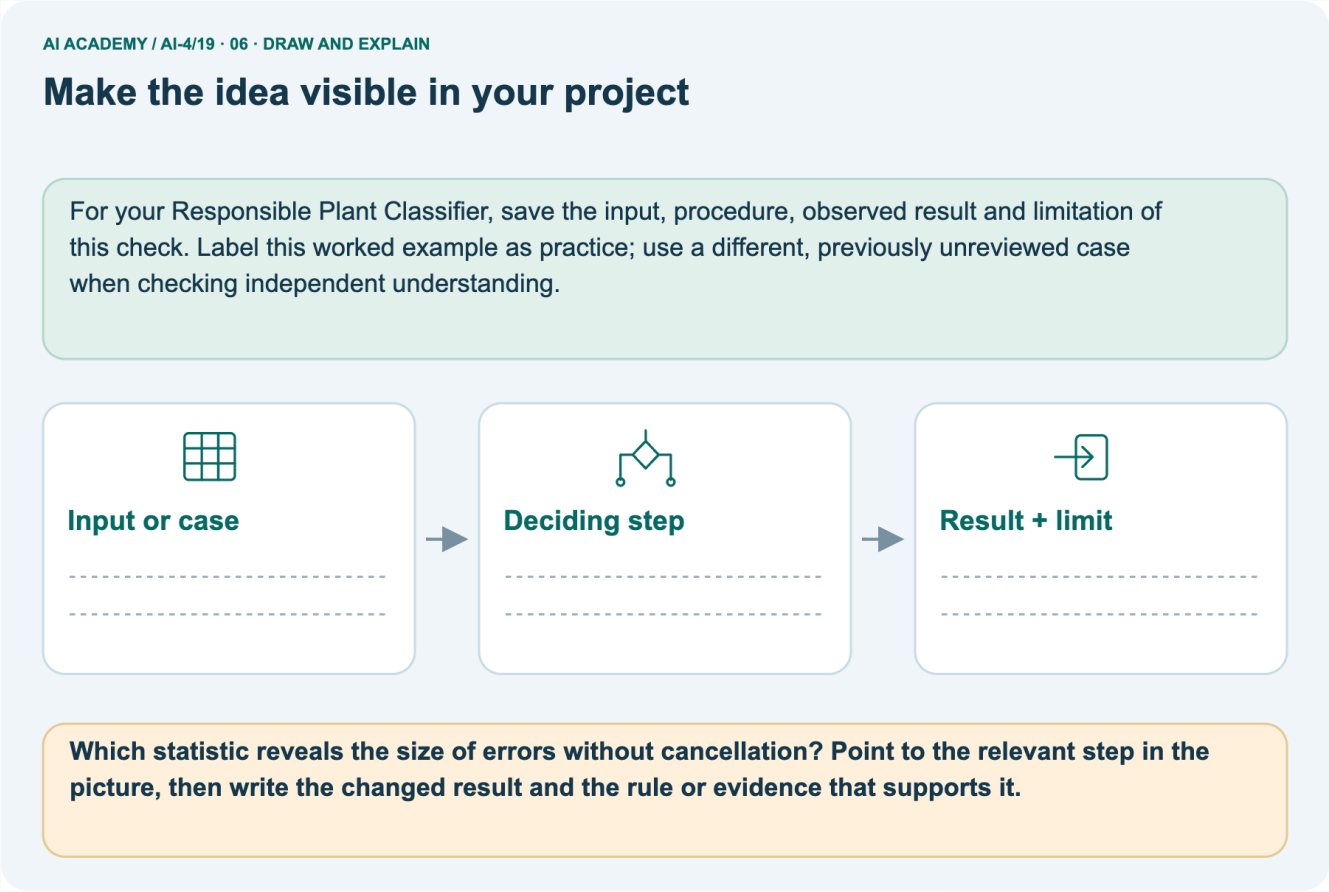
Read the picture in order. Mean **residual** is 1/3 cm; MAE is 5/3 cm, so small signed mean does not imply small errors. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Always declare the **residual** sign convention; some software uses predicted minus observed.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which statistic reveals the size of errors without cancellation? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Regression and Residuals: the case

02. Follow the deciding steps

03. Observed minus predicted height

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.